

deserts_by_race_analysis_2

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Set up

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.2      v tibble    3.3.1
## v lubridate   1.9.5      v tidyr     1.3.2
## v purrr       1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
desert_threshold <- 0.33
```

```
# Loading csv files
```

```
race_data <- read_csv("pop_hispanic_blacknh.csv")
```

```
## Rows: 84781 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (2): GEO_ID, NAME
## dbl (5): total_pop, hispanic, black_nonh, pct_hispanic, pct_black_nonh
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
childcare_data <- read_csv("cap_ccaccess_2025_revised_02282026.csv")
```

```
## Rows: 2281139 Columns: 14
## -- Column specification -----
## Delimiter: ","
## chr (7): state_fips, state_name, state_abbrev, county_fips, county_name, scho...
## dbl (7): longitude, latitude, adj_supply, adj_supply_fcc, adj_supply_ccc, ad...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
joined_rural <- read_csv("joined_rural.csv")
```

```
## Rows: 2281139 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (9): state_name, state_abbrv, county_fips, county_name, schooldistrict_c...
## dbl (9): longitude, latitude, adj_supply, adj_supply_fcc, adj_supply_ccc, ad...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Data cleaning

```
# Cleaning the race data
race_data <- read_csv("pop_hispanic_blacknh.csv") %>%
  # Strip the "1400000US" prefix to get the 11-digit census tract FIPS
  mutate(county_fips = str_sub(str_remove(GEO_ID, "1400000US"), 1, 5)) %>%
  filter(!is.na(pct_hispanic), !is.na(pct_black_nonh)) %>%
  group_by(county_fips) %>%
  summarise(
    avg_pct_hispanic = weighted.mean(pct_hispanic, w = total_pop, na.rm = TRUE),
    avg_pct_black_nonh = weighted.mean(pct_black_nonh, w = total_pop, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## Rows: 84781 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (2): GEO_ID, NAME
## dbl (5): total_pop, hispanic, black_nonh, pct_hispanic, pct_black_nonh
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Joining with childcare data
childcare_data <- read_csv("cap_ccaccess_2025_revised_02282026.csv")
```

```
## Rows: 2281139 Columns: 14
## -- Column specification -----
## Delimiter: ","
## chr (7): state_fips, state_name, state_abbrv, county_fips, county_name, scho...
## dbl (7): longitude, latitude, adj_supply, adj_supply_fcc, adj_supply_ccc, ad...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
childcare_with_race <- childcare_data %>%
  mutate(county_fips = str_pad(as.character(county_fips), width = 5, pad = "0")) %>%
  left_join(race_data, by = "county_fips") %>%
  mutate(desert = adj_supply <= desert_threshold)
```

```

# Check join quality
cat("Rows with Hispanic data:", sum(!is.na(childcare_with_race$avg_pct_hispanic)), "\n")

## Rows with Hispanic data: 2259432

cat("Rows missing Hispanic data:", sum(is.na(childcare_with_race$avg_pct_hispanic)), "\n")

## Rows missing Hispanic data: 21707

# Create rural flag
rural_lookup <- joined_rural %>%
  select(longitude, latitude, rural) %>%
  distinct()

# Joining with rural flag
childcare_with_race_rural <- childcare_with_race %>%
  left_join(rural_lookup, by = c("longitude", "latitude"))

# Check join quality
cat("Rows with rural status:", sum(!is.na(childcare_with_race_rural$rural)), "\n") ## Should be 2277112

## Rows with rural status: 2277112

cat("Rows missing rural status:", sum(is.na(childcare_with_race_rural$rural)), "\n") ## Should be 4027

## Rows missing rural status: 4027

# Remove missing values and code new variables
childcare_with_race_rural <- childcare_with_race_rural %>%
  filter(!is.na(adj_supply), !is.na(avg_pct_hispanic), !is.na(avg_pct_black_nonh), !is.na(rural)) %>%
  mutate(
    desert = adj_supply <= desert_threshold,
    county_type = case_when(
      avg_pct_hispanic > 50 ~ "Majority Hispanic",
      avg_pct_black_nonh > 50 ~ "Majority Black NH",
      TRUE ~ "Neither majority"
    )
  )

```

State aggregated data

```

# Creating state-level dataset
state_race_desert <- childcare_with_race_rural %>%
  filter(!is.na(avg_pct_hispanic), !is.na(avg_pct_black_nonh), !is.na(adj_supply)) %>%
  group_by(state_name) %>%
  summarise(
    pct_in_desert = round(mean(desert) * 100, 1),
    avg_pct_hispanic = round(mean(avg_pct_hispanic), 1),
    avg_pct_black_nonh = round(mean(avg_pct_black_nonh), 1),
    .groups = "drop"
  )

```

Comparing desert rates by majority-minority county composition

The values changed slightly: 52.9 -> 52.2 45.8 -> 45.8 34.6 -> 34.6

```
# Creating the tibble
childcare_with_race_rural %>%
  group_by(county_type) %>%
  summarise(
    total_areas = n(),
    pct_in_desert = round(mean(desert) * 100, 1),
    .groups = "drop"
  ) %>%
  arrange(desc(pct_in_desert))
```

```
## # A tibble: 3 x 3
##   county_type      total_areas pct_in_desert
##   <chr>          <int>      <dbl>
## 1 Majority Hispanic    150746      52.2
## 2 Neither majority    2055657     45.8
## 3 Majority Black NH    49002      34.6
```

Comparing desert rates by majority-minority county composition and urbanicity

No change in values (expected).

```
# Creating the tibble
childcare_with_race_rural %>%
  filter(!is.na(adj_supply), !is.na(avg_pct_hispanic), !is.na(avg_pct_black_nonh), !is.na(rural)) %>%
  mutate(
    desert = adj_supply <= desert_threshold,
    county_type = case_when(
      avg_pct_hispanic >= 50 ~ "Majority Hispanic",
      avg_pct_black_nonh >= 50 ~ "Majority Black NH",
      TRUE ~ "Neither majority"
    )
  ) %>%
  group_by(county_type, rural) %>%
  summarise(
    total_areas = n(),
    pct_in_desert = round(mean(desert) * 100, 1),
    .groups = "drop"
  ) %>%
  arrange(county_type, rural)
```

```
## # A tibble: 6 x 4
##   county_type      rural total_areas pct_in_desert
##   <chr>          <dbl>      <int>      <dbl>
## 1 Majority Black NH     0      40064      29.8
## 2 Majority Black NH     1       8938      55.7
## 3 Majority Hispanic     0     138460      51.5
## 4 Majority Hispanic     1      12286      60.2
## 5 Neither majority     0    1723969       42
## 6 Neither majority     1     331688      65.6
```

States driving rural majority-hispanic and rural majority-Black desert rates

There were also no changes here (expected).

```
# Make the tibble
state_race_rural_desert <- childcare_with_race_rural %>%
  filter(
    !is.na(adj_supply), !is.na(avg_pct_hispanic),
    !is.na(avg_pct_black_nonh), !is.na(rural)
  ) %>%
  mutate(
    desert = adj_supply <= desert_threshold,
    county_type = case_when(
      avg_pct_hispanic >= 50 ~ "Majority Hispanic",
      avg_pct_black_nonh >= 50 ~ "Majority Black NH",
      TRUE ~ "Neither majority"
    )
  ) %>%
  filter(rural == 1, county_type != "Neither majority") %>%
  group_by(state_name, county_type) %>%
  summarise(
    total_areas = n(),
    pct_in_desert = round(mean(desert) * 100, 1),
    .groups = "drop"
  )
print(state_race_rural_desert)
```

```
## # A tibble: 20 x 4
##   state_name    county_type    total_areas pct_in_desert
##   <chr>         <chr>          <int>         <dbl>
## 1 Alabama      Majority Black NH    1002          64
## 2 Arizona      Majority Hispanic     346         64.7
## 3 Arkansas     Majority Black NH     905         48.5
## 4 California   Majority Hispanic     661         57.5
## 5 Colorado     Majority Hispanic      75         86.7
## 6 Florida      Majority Black NH     307         58.6
## 7 Florida      Majority Hispanic     337         40.1
## 8 Georgia      Majority Black NH     875         66.6
## 9 Kansas       Majority Hispanic     991         81.1
## 10 Louisiana   Majority Black NH     193         53.4
## 11 Mississippi Majority Black NH   3092         48.3
## 12 New Mexico  Majority Hispanic   2557         49.5
## 13 North Carolina Majority Black NH     624         57.7
## 14 Oklahoma    Majority Hispanic     179         97.2
## 15 Puerto Rico Majority Hispanic     945        100
## 16 South Carolina Majority Black NH   1242         62.9
## 17 Tennessee   Majority Black NH     118         74.6
## 18 Texas       Majority Hispanic   5979         54.9
## 19 Virginia    Majority Black NH     580         54.1
## 20 Washington  Majority Hispanic     216          56
```

```
# Rural majority-hispanic by state
hispanic_states <- state_race_rural_desert %>%
```

```
filter(county_type == "Majority Hispanic") %>%  
  arrange(desc(pct_in_desert))  
  
# Rural majority black_nh by state  
black_states <- state_race_rural_desert %>%  
  filter(county_type == "Majority Black NH") %>%  
  arrange(desc(pct_in_desert))
```